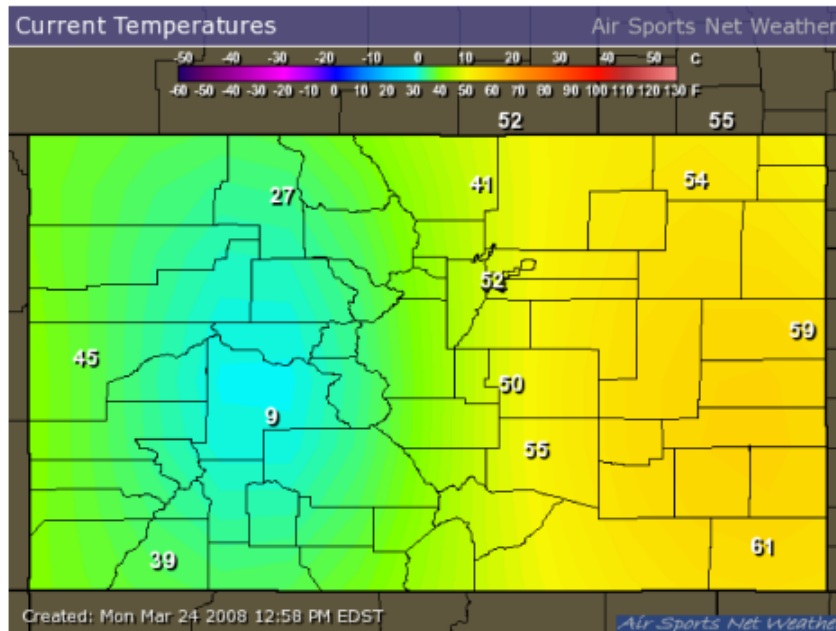


Math 2551 Worksheet: Applications of Double Integrals

- The figure below shows a temperature map of Colorado. Use the data to estimate the average temperature in the state using 4, 16, and 25 subdivisions. Give both an upper and lower estimate for each number of subdivisions. Why do we like Colorado for this problem? What other state(s) or countries might we like?



- Which do you think will be larger, the average value of $f(x, y) = xy$ over the square $0 \leq x \leq 1, 0 \leq y \leq 1$, or the average value of f over the quarter circle $x^2 + y^2 \leq 1$ in the first quadrant? Calculate them to find out.
- If $f(x, y) = 100(y + 1)$ represents the population density in people per square mile of a planar region on Earth, where x and y are measured in miles, find the number of people in the region bounded by the curves $x = y^2$ and $x = 2y - y^2$.